

MUHAMMAD ABDULWADUD AYINDE

Software Engineer || Machine Learning Researcher

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 [LinkedIn Profile](#) |  [GitHub Profile](#) |  [Google Scholar Profile](#) | Ilorin, Kwara State, Nigeria

Research Interests

- Interpretable and Uncertainty-Aware Machine Learning
- Software Defect Prediction and Mining Software Repositories
- Digital Health AI: Disease Risk Prediction, Diagnosis Support, Clinical Decision-Making
- Deep Learning Architectures for Tabular Data (TabNet)

Education

Kwara State University (Kwasu), Malete, Nigeria

Sep 2022 – Jul 2026

B.Sc. in Computer Science, Second Class Upper Division (CGPA 4.01/5.00)

- Relevant Courses: Data Structures & Algorithms, Artificial Intelligence & Expert Systems, Software Engineering, Database Management Systems (DBMS), Operating Systems, Object-Oriented Programming, Data Communication & Networks.

Research Experience

Undergraduate Researcher, Software Defect Prediction Analysis

2026 – Present

Kwara State University, Malete, Nigeria

- Designed and implemented a proactive Software Defect Prediction (SDP) system using the TabNet deep learning architecture on the NASA CM1 benchmark dataset.
- Addressed class imbalance by applying the Synthetic Minority Over-sampling Technique (SMOTE) to the training partition, balancing the dataset from 562 clean vs. 43 defective modules.
- Leveraged TabNet's sequential attention mechanism for native feature selection, identifying LOC_CODE, LOC_BLANK, and PERCENT-based metrics as the top predictive features without post-hoc approximation tools.
- Achieved a 95% recall rate (F1-score: 0.6032), verified via confusion matrix (19 true positives, 1 false negative), and deployed a live diagnostic dashboard (software-def.onrender.com) translating model attention weights into actionable Z-score deviations.
- Collaborated with Abiola Akorede and Abdul-Rasheed Adeniyi Babalola under the supervision of Mr. H. B. Sulaiman.

AI Developer, Build with Gemma: GenAI for SDGs Hackathon (Kaggle/Kwasu)

2026

- Engineered MamaCare-Triage, an offline-capable, multimodal, AI-driven health triage assistant for rural clinics, using Gemma 4's multimodal reasoning and native function-calling to return a condition/urgency assessment, a safe medication dosage, and danger-sign alerts, translated into Yoruba for CHWs, directly addressing UN SDG 3 (Good Health and Well-being).

Top 10 Global Finalist, Aspire × Cayu Global AI Hackathon

Jan 2026

- Designed and engineered an AI-powered Community Knowledge Sharing platform to bridge localized information gaps, demonstrating rapid prototyping of scalable AI solutions under strict time constraints.

Software Projects

GlobalCoachAI, Intelligent Career Orchestration Platform

2026

- Built a full-stack career-coaching platform (React.js, Vite, Tailwind CSS, Node.js, Express.js, PostgreSQL hosted on Neon, Drizzle ORM) with an asynchronous engine that compares user CVs against live market demand.
- Integrated Google Gemini 1.5 Flash for semantic job-matching and built a data-driven dashboard (TanStack Query, interactive charts) tracking application activity and skill readiness.

CropDx, Field-First Crop Disease Detection Assistant

2026

- Built a full-stack diagnostic tool (React/Vite frontend, Flask API) using a classical machine learning pipeline (HSV, GLCM, and OpenCV feature extraction with a calibrated RBF SVM), deliberately chosen over a CNN for inspectability on low-power field devices.
- Designed a SQLite-backed disease and recommendation database and a free, multilingual (10-language) device-voice guide requiring no AI credits, for use in low-connectivity field conditions.

Teaching Experience

Peer Tutor, Department of Computer Science

2023 – 2026

Kwara State University, Malete, Nigeria

- Conducted tutorials in CSC 201 (Application of Computer Technology) and CSC 305 (Data Structures and Algorithms) for undergraduate students.
- Facilitated problem-solving and live coding sessions, enhancing students' understanding of programming logic and computer architecture.

Mathematics and Computer Tutor, Al-Furqan Group of Schools

2023 – Present

Ilorin, Nigeria

- Delivered Mathematics and Computer Studies tutorials for senior students preparing for JAMB and SSCE examinations.
- Mentored students to strengthen quantitative reasoning, digital literacy, and scientific problem-solving skills.

Data Analysis Facilitator, NAICTS SRC “Alternative to Coding” Bootcamp

Jun 22–24, 2026

Kwara State University (KWASU), Malete, Nigeria

- Facilitated the Data Analysis track of a 3-day practical bootcamp organized by the National Association of Information and Communication Technology Students (NAICTS) Student Representative Council, introducing students to data analysis fundamentals.

Industrial Experience

Software Engineer, Pacific Artis

Sep 2026 – Present

Mawson Lakes, South Australia, Australia (Remote)

- Working as a Software Engineer at Pacific Artis, contributing to AI-driven maintenance and reliability engineering initiatives.

Backend Development Intern, BitsPro Consulting Limited

Aug 2026 – Present

United Kingdom (Remote)

- Designing and implementing RESTful CRUD APIs in Node.js and Express, routes, middleware, and controllers, as part of a 26-week AI-integrated internship building a full ecommerce API from the ground up, with weekly mentor-reviewed deliverables and AI-assisted workflows (Copilot, Claude).
- Set up and deployed a local Express server with a working health-check endpoint, and independently designed and documented a full Products CRUD API (endpoints, schema, validation logic) before implementation.

Software Engineering Intern, SystemSpecs (Remita)

Jun 2024 – 2026

Lagos, Nigeria

- Collaborated on designing and testing backend infrastructure and RESTful APIs using TypeScript/Node.js within an Agile team environment.
- Optimized responsive client-facing UI components using React.js and Tailwind CSS.

Content Intern, Trayce (AI Context-Layer Startup)

Apr 2026 – Present

Remote

- Produced developer-focused content, translating complex technical features into plain-language summaries and actionable insights.

Professional Memberships

- Cowrywise Campus Ambassador (Dec 2025 – Present), promoting financial literacy and fintech awareness among students.
- Internshala Student Partner
- Global Freelancers Union (GFU), Member
- Volunteer Tech Community Member: GDG Ilorin, Hult Prize at KWASU, Malete Tech Forum

Certifications

- Implementing Ethical AI in Academic Research and Writing, RUDN University Professional Scholarship Program (2026)

Leadership Positions

- Chairman, Executive Planning Committee, KWASU NACOS Tech Conference (2026)
- Facilitator & Coordination Team, BuildWithAI Event & Future Flip, Al-Hikmah University
- Kectil Youth Leadership Fellow, The Malmar Knowles Family Foundation (Jan 2026 – Present)
- Certified Aspire Leader, Aspire Institute, founded by Harvard Faculty (Oct 2025)

Voluntary / Community Service

Election Campaign Monitoring Volunteer

2026 – 2027

2027 General Election Campaign Monitoring Project

Center for Peace and Strategic Studies, University of Ilorin, in collaboration with CRDAMES & RUDN University

- Monitor political campaign activities across mass and social media platforms.
- Collect and organize media and social media data for research and analysis.
- Contribute to data-driven reports and summaries on campaign activities.
- Support the development and dissemination of research communication content.
- Collaborate with academic and research teams on election monitoring activities.
- Volunteer, GDG Ilorin, Hult Prize at KWASU, Malete Tech Forum

Conferences Attended

- KWASU NACOS Tech Conference: “Advancing Innovation Through Technology and Collaboration”, Kwara State University, 2026
- KWASU Tech Festival (KTF), Kwara State University, 2026
- Kwara AI Summit, Ilorin Innovation Hub, Kwara State, 2025
- Kwarabuild Tech Conference (KBTC), Ilorin, Kwara State, 2025
- On-Campus Startup Workshop: “From Idea to Market-Ready Product” (Hult Prize), Fars Tech Hub, KWASU, 14 Dec 2025
- Sulh Summit: “Towards Harmonised Classical Islamic Jurisprudence and Statute”, Kwara State University, 20 Jul 2026